

Week 1 Mandatory SQL Hands-on Answers

Exercise 1: Ranking and Window Functions

Goal: Find the top 3 most expensive products in each category using ROW_NUMBER(), RANK(), and DENSE_RANK().

```
-- Using ROW_NUMBER()
SELECT *
FROM (
    SELECT ProductID, ProductName, Category, Price,
           ROW_NUMBER() OVER(PARTITION BY Category ORDER BY Price DESC)
    AS RowNum
    FROM Products
) AS RankedProducts
WHERE RowNum <= 3;
```

```
-- Using RANK()
SELECT ProductID, ProductName, Category, Price,
       RANK() OVER(PARTITION BY Category ORDER BY Price DESC) AS
ProductRank
FROM Products;
```

```
-- Using DENSE_RANK()
SELECT ProductID, ProductName, Category, Price,
       DENSE_RANK() OVER(PARTITION BY Category ORDER BY Price DESC) AS
DenseRank
FROM Products;
```

Exercise 1: Create a Stored Procedure

Create a stored procedure to retrieve employee details by department.

```
CREATE PROCEDURE sp_GetEmployeesByDepartment
    @DepartmentID INT
AS
BEGIN
    SELECT EmployeeID, FirstName, LastName, DepartmentID, Salary,
    JoinDate
    FROM Employees
    WHERE DepartmentID = @DepartmentID;
END;
GO
```

```
EXEC sp_GetEmployeesByDepartment 1;
```

Exercise 5: Return Data from a Stored Procedure

Create a stored procedure that returns the total number of employees in a department.

```
CREATE PROCEDURE sp_GetEmployeeCount
    @DepartmentID INT
AS
BEGIN
    SELECT COUNT(*) AS TotalEmployees
    FROM Employees
    WHERE DepartmentID = @DepartmentID;
END;
GO

EXEC sp_GetEmployeeCount 3;
```